

SUMMER INTERNSHIP PROJECT REPORT

Early Warning System for Identifying Financially Distressed Indian Publicly Traded Companies

Submitted by

Ashis Pal

M.A. Applied Quantitative Finance
Madras School of Economics

Project Guide

Koulika Paul

Period of Internship:

18th May 2026 – 31st July 2026

Report submitted to:

IDEAS – Institute of Data Engineering, Analytics and Science Foundation
Indian Statistical Institute, Kolkata

July 2026

Contents

1	Acknowledgement	1
2	Abstract	2
3	Introduction	4
4	Objective of the Study	6
5	Methodology	8
5.1	Overview of the Methodology	8
5.2	Data Collection	9
5.3	Data Cleaning and Pre-processing	10
5.4	Target Variable Construction	11
5.5	Feature Engineering	11
5.6	Exploratory Data Analysis (EDA)	13
5.7	Feature Selection	13
5.8	Data Preparation for Machine Learning	14
5.9	Model Development	14
5.10	Model Evaluation	15
5.11	Hyperparameter Optimization	15
5.12	Model Interpretation	16
5.13	Threshold Optimization	16
5.14	Deployment Demonstration	16
6	Data Analysis and Results	17
6.1	Dataset Overview	17
6.2	Data Quality Assessment	18
6.2.1	Duplicate Analysis	18
6.2.2	Missing Value Analysis	18
6.3	Target Variable Analysis	19
6.4	Exploratory Data Analysis	20
6.4.1	Exploratory Analysis of Raw Financial Variables	20

6.4.2	Exploratory Analysis of Engineered Financial Ratios	23
6.5	Feature Selection	24
6.6	Performance of Machine Learning Models	26
6.6.1	Baseline Model Performance	27
6.6.2	Hyperparameter Tuning	28
6.6.3	Performance of Optimized Models	29
6.6.4	Model Selection	30
6.6.5	Interpretation of the Model via SHAP	30
6.6.6	Threshold Optimization	33
6.6.7	Final Summary	35
6.7	Early Warning Dashboard	37
7	Conclusion	39
7.1	Conclusions	39
7.2	Future Scope	40
	Appendices	41

Chapter 1

Acknowledgement

I would like to express my sincere gratitude to the Institute of Data Engineering, Analytics and Science Foundation (IDEAS) for providing me with the opportunity to participate in the Summer Internship Program 2026. This internship has been an invaluable learning experience and has greatly enhanced my understanding of data analysis, statistical modelling, and machine learning techniques.

I am deeply thankful to my mentor, Ms. Koulika Paul, for her constant guidance, encouragement, and support throughout the project. Her valuable suggestions and insightful feedback helped me successfully complete this work.

I would also like to extend my heartfelt gratitude to the faculty members of Madras School of Economics, Chennai for providing me with a strong academic foundation and continuous motivation to pursue statistical research and data science applications.

Finally, I would like to thank my family and friends for their unwavering support, encouragement, and motivation throughout the internship period.

Ashis Pal

M.A. Applied Quantitative Finance

Madras School of Economics

Chapter 2

Abstract

Financial distress prediction plays a vital role in corporate risk management by enabling the early identification of financially vulnerable firms before severe financial deterioration occurs. This study develops an Early Warning System (EWS) for predicting the future financial distress of Indian publicly listed companies using accounting-based financial indicators and machine learning techniques. Financial statement data were collected from companies listed in the Nifty 500 Index, followed by comprehensive data preprocessing, feature engineering, missing value treatment, and the construction of a future distress target variable. To address the class imbalance problem, the Synthetic Minority Over-sampling Technique for Nominal and Continuous variables (SMOTENC) was employed.

A systematic feature selection framework incorporating the Mann–Whitney U test, Spearman correlation analysis, Variance Inflation Factor (VIF) analysis, and Least Absolute Shrinkage and Selection Operator (LASSO) validation was implemented to identify the most informative financial predictors while reducing redundancy and multicollinearity. Six machine learning algorithms—Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, and CatBoost—were evaluated using multiple performance metrics, with Precision–Recall Area Under the Curve (PR–AUC) adopted as the primary evaluation metric due to the imbalanced nature of the dataset. Following hyperparameter optimization using Optuna, the Optimized Random Forest model achieved the best overall performance, recording a PR–AUC of 0.752, ROC–AUC of 0.988, and an F1-score of 0.729.

Model interpretability was enhanced using SHapley Additive exPlanations (SHAP), which identified Interest Coverage, Return on Assets (ROA), Net Profit, Operating Profit, and Equity Ratio as the most influential predictors of financial distress. Threshold optimization further improved the balance between precision and recall by selecting an optimal classification threshold of 0.535. Finally, an Early Warning Dashboard was developed to rank companies according to their predicted distress probabilities and generate forward-looking predictions using the latest available financial statements.

The proposed framework demonstrates that integrating accounting-based financial

analysis with explainable machine learning provides an accurate, interpretable, and practical decision-support system for investors, lenders, regulators, and corporate managers to identify financial distress at an early stage and facilitate timely risk management.

Keywords: Financial Distress Prediction, Early Warning System, Machine Learning, Random Forest, Explainable Artificial Intelligence (XAI), SHAP, SMOTENC, Financial Statement Analysis, Corporate Risk Assessment.

Chapter 3

Introduction

Financial distress is a critical issue for businesses, investors, creditors, and regulatory bodies, as it is often a precursor to business failure or bankruptcy. Companies experiencing financial distress typically exhibit declining financial performance, persistent losses, high leverage, inadequate liquidity, and an inability to meet their financial obligations. Identifying such companies at an early stage is essential because timely intervention can reduce financial losses, improve investment decisions, and strengthen risk management practices.

Traditionally, financial distress has been predicted using statistical techniques such as the Altman Z-Score, Ohlson O-Score, and logistic regression, which estimate the likelihood of corporate failure based on financial ratios. Although these methods are widely used because of their simplicity and interpretability, they generally assume linear relationships between financial variables and the probability of distress. Recent advances in machine learning have enabled the development of more sophisticated predictive models capable of capturing complex nonlinear relationships and interactions among financial indicators, thereby improving predictive performance.

Despite the availability of several financial distress prediction models, publicly available data for Indian listed companies rarely contain explicit distress labels, making supervised machine learning challenging. Furthermore, many traditional approaches focus primarily on bankruptcy rather than identifying financial distress at an earlier stage. This project addresses these challenges by developing an Early Warning System capable of identifying Indian publicly traded companies that are likely to experience financial distress before conventional warning signs become evident. Such an approach enables stakeholders to undertake timely preventive actions.

The dataset used in this study was constructed from the financial statements of Indian publicly listed companies collected from Screener.in and other publicly available sources through web scraping. Since no explicit distress labels were available, a rule-based target variable was developed using accounting-based criteria, including negative net worth, excessive leverage combined with negative operating cash flows, and low interest cover-

age. The dataset subsequently underwent data cleaning, feature engineering, exploratory data analysis, statistical feature selection, missing value treatment, class imbalance handling, and machine learning model development. Multiple classification algorithms were trained and evaluated using stratified cross-validation and Bayesian hyperparameter optimization with Optuna. To improve model transparency, SHapley Additive exPlanations (SHAP) were employed for model interpretation, followed by threshold optimization and the development of an Early Warning Dashboard for practical deployment.

The outcome of this study is an interpretable Early Warning System that predicts the likelihood of financial distress using publicly available financial statement information. Beyond achieving strong predictive performance, the proposed framework emphasizes model interpretability and practical applicability, providing a valuable decision-support tool for investors, lenders, regulators, and corporate managers.

Chapter 4

Objective of the Study

The primary objective of this study is to develop an interpretable Early Warning System capable of predicting the future financial distress of Indian publicly listed companies using accounting-based financial indicators and machine learning techniques.

The specific objectives of the study are as follows:

1. To collect and construct a comprehensive financial dataset for Indian publicly listed companies using publicly available financial statement information.
2. To develop an accounting-based target variable for financial distress prediction in the absence of explicit bankruptcy or distress labels.
3. To preprocess the dataset through data cleaning, missing value treatment, and feature engineering to improve data quality and model performance.
4. To identify the most relevant financial predictors using statistical feature selection techniques, including the Mann–Whitney U test, correlation analysis, Variance Inflation Factor (VIF), and LASSO validation.
5. To address class imbalance using the Synthetic Minority Oversampling Technique for Nominal and Continuous variables (SMOTENC).
6. To develop and compare multiple machine learning models for financial distress prediction.
7. To optimize the best-performing model using Bayesian hyperparameter optimization with Optuna.
8. To interpret the final prediction model using SHapley Additive exPlanations (SHAP) for improved model transparency.
9. To determine an optimal probability threshold for financial distress classification.

10. To develop an Early Warning Dashboard capable of ranking companies according to their predicted financial distress probabilities for practical decision support

Hypothesis Tested

As part of the feature selection process, the study tested whether the distributions of selected financial indicators differed significantly between healthy and financially distressed companies using the *Mann–Whitney U Test*.

The hypotheses were:

1. Null Hypothesis (H_0): The distribution of a financial indicator is the same for healthy and financially distressed companies.
2. Alternative Hypothesis (H_1): The distribution of a financial indicator differs significantly between healthy and financially distressed companies.

Financial indicators with statistically significant differences were considered for further analysis along with correlation analysis and multicollinearity assessment and also a comparative analysis with Lasso Regression as well.

Chapter 5

Methodology

5.1 Overview of the Methodology

The primary objective of this study is to develop an **Early Warning System (EWS)** capable of predicting the future financial distress of Indian publicly listed companies using accounting-based financial information and machine learning techniques. The proposed framework integrates data collection, preprocessing, feature engineering, statistical feature selection, predictive modelling, model interpretation, and threshold optimization into a unified and interpretable financial distress prediction system.

The study begins with the collection of financial statement information for companies listed in the **Nifty 500 Index**. Financial data were obtained through web scraping from publicly available sources and consolidated into a panel dataset containing multiple years of financial information for each company. Since the dataset did not contain explicit financial distress labels, an accounting-based target variable was constructed using predefined financial distress criteria derived from financial theory.

Following data collection, the dataset underwent extensive preprocessing to improve data quality and modelling performance. This stage included duplicate removal, missing value treatment, feature engineering, and the computation of financial ratios that capture different dimensions of corporate financial health, including profitability, liquidity, leverage, solvency, and operational efficiency.

To identify the most informative predictors while minimizing redundancy and multicollinearity, a systematic feature selection framework was adopted. The framework combined the **Mann–Whitney U test**, Spearman correlation analysis, **Variance Inflation Factor (VIF)** analysis, and **Least Absolute Shrinkage and Selection Operator (LASSO)** validation. This multi-stage approach ensured that the final feature set remained both statistically significant and financially interpretable.

As financial distress prediction inherently involves an imbalanced classification problem, the **Synthetic Minority Oversampling Technique for Nominal and Contin-**

uous variables (SMOTENC) was employed to balance the training dataset prior to model development. Subsequently, six supervised machine learning algorithms—Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, and CatBoost—were trained and evaluated using stratified cross-validation. Hyperparameter optimization was performed using the **Optuna** framework to improve the predictive performance of the best-performing models.

Considering the imbalanced nature of the dataset, model performance was primarily evaluated using the **Precision–Recall Area Under the Curve (PR–AUC)**, while additional metrics such as Accuracy, Precision, Recall, F1-score, and ROC–AUC were also reported to provide a comprehensive assessment of predictive performance.

To enhance transparency and interpretability, the final prediction model was analyzed using **SHapley Additive exPlanations (SHAP)**, enabling the contribution of individual financial variables toward distress prediction to be quantified and visualized. Furthermore, probability threshold optimization was performed to determine the classification threshold that achieved the best balance between Precision and Recall.

Finally, the optimized prediction model was deployed as an **Early Warning Dashboard** capable of estimating the probability of future financial distress and ranking companies according to their predicted financial risk. The overall methodological framework adopted in this study is presented in Figure ??.

5.2 Data Collection

The dataset was constructed using publicly available financial information of Indian publicly traded companies.

Data Sources

- Official **NIFTY 500** company list obtained from the National Stock Exchange (NSE) through web scraping.
- Historical financial statement data collected from *Screener.in*.
- Sector information collected from *Yahoo Finance (yfinance)* using web scraping.

Dataset Characteristics

- Panel dataset containing multiple financial years for each company.
- Financial information collected from the **Balance Sheet**, **Profit and Loss Statement**, and **Cash Flow Statement**.
- Included both raw financial variables and derived financial ratios.

Examples of Variables Collected

- Sales
- Net Profit
- Operating Profit
- Borrowings
- Total Assets
- Total Liabilities
- Equity Capital
- Reserves
- Cash Flow from Operations (CFO)
- Free Cash Flow (FCF)
- Interest Expense
- Fixed Assets
- Investments
- Depreciation
- Other Assets
- Other Liabilities

5.3 Data Cleaning and Pre-processing

The collected dataset was examined for inconsistencies before model development.

The following preprocessing steps were performed:

- Removed duplicate observations.
- Checked data types of all variables.
- Assessed missing values for every feature.
- Treated missing values using appropriate imputation methods based on the characteristics of each variable (*forward fill, median imputation, mode imputation, and zero imputation*).

- Standardized company identifiers and financial years.
- Verified the consistency of financial records across companies.

These preprocessing steps ensured that the dataset was clean, consistent, and suitable for subsequent feature engineering and machine learning model development.

5.4 Target Variable Construction

Since publicly available datasets do not provide an explicit financial distress label, a **rule-based target variable** was developed.

A company was classified as financially distressed if **any** of the following conditions were satisfied:

- Negative Net Worth

Shareholder equity less than zero.

- Extremely high leverage combined with persistent negative Cash Flow from Operations (CFO).

Debt-to-Assets Ratio $\geq$ Sector 90th Percentile and negative CFO for two consecutive years.

- *Interest Coverage Ratio remaining below one for two consecutive years.*

To develop an **Early Warning System** rather than a contemporaneous classification model, the distress label was shifted one financial year ahead. Consequently, the financial information from year t was used to predict the financial distress status in year $t + 1$.

5.5 Feature Engineering

To better capture different dimensions of a company's financial health, several financial ratios were derived from the original financial statement variables.

(a) Profitability Indicators

The profitability indicators used in this study are presented in Table 5.1.

Table 5.1: Profitability Indicators

Feature	Formula
Return on Assets (ROA)	Net Profit / Total Assets
Return on Equity (ROE)	Net Profit / Equity
Operating Margin	Operating Profit / Sales
Net Profit Margin	Net Profit / Sales

(b) Leverage and Capital Structure Indicators

The leverage and capital structure indicators used in this study are presented in Table 5.2.

Table 5.2: Leverage and Capital Structure Indicators

Feature	Formula
Debt-to-Equity Ratio	Borrowings / Equity
Equity Ratio	Equity / Total Assets

(c) Liquidity and Cash Flow Indicators

The liquidity and cash flow indicators used in this study are presented in Table 5.3.

Table 5.3: Liquidity and Cash Flow Indicators

Feature	Formula
Operating Cash Flow to Assets	Cash Flow from Operations / Total Assets
Free Cash Flow to Assets	Free Cash Flow / Total Assets

(d) Asset Structure and Efficiency Indicators

The asset structure and efficiency indicators used in this study are presented in Table 5.4.

Table 5.4: Asset Structure and Efficiency Indicators

Feature	Formula
Asset Turnover	Sales / Total Assets
Fixed Asset Ratio	Fixed Assets / Total Assets
Capital Work-in-Progress (CWIP) Ratio	CWIP / Total Assets
Investment Ratio	Investments / Total Assets

5.6 Exploratory Data Analysis (EDA)

Exploratory analysis was performed to understand the characteristics of the dataset before model development.

The analysis included:

- Descriptive statistics.
- Distribution of financial variables.
- Missing value analysis.
- Correlation analysis.
- Class distribution.

The insights obtained from **EDA** guided subsequent feature selection and preprocessing decisions.

5.7 Feature Selection

A **multi-stage feature selection strategy** was adopted to identify informative predictors while reducing redundancy. The procedure consisted of:

1. **Mann–Whitney U Test**

Used to identify variables that differed significantly between healthy and distressed companies.

2. **Correlation Analysis**

Highly correlated variables were identified to reduce redundancy.

3. **Variance Inflation Factor (VIF)**

Used to detect multicollinearity among predictors.

4. **Financial Interpretation**

The final feature set was selected by considering both statistical significance and the financial relevance of the variables.

5. **LASSO Logistic Regression (Robustness Check)**

Explored as an alternative embedded feature selection technique.

The features selected by LASSO showed substantial agreement with those identified through the statistical feature selection framework.

LASSO retained most of the statistically selected predictors, discarding only a small number of features (2).

As the resulting feature subset was largely consistent with the original selection, the statistically derived feature set was retained for the subsequent analysis to maintain methodological consistency.

5.8 Data Preparation for Machine Learning

Before model training, the dataset was prepared using the following steps:

- Train-test split using stratified sampling.
- Missing value imputation.
- Feature standardization where appropriate.
- One-Hot Encoding of categorical variables.
- Treatment of class imbalance using **SMOTENC** on the training dataset only.

Performing preprocessing after the train-test split helped prevent **data leakage** and ensured unbiased model evaluation.

5.9 Model Development

Six supervised machine learning algorithms were developed and compared.

The models considered were:

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost
- LightGBM
- CatBoost

Each model was trained using the same training dataset to enable a fair performance comparison.

5.10 Model Evaluation

The performance of each model was evaluated using multiple classification metrics.

The evaluation metrics included:

- Precision
- Recall
- F1-score
- ROC-AUC
- Precision-Recall Area Under the Curve (PR-AUC)

Among these, **PR-AUC** was selected as the primary evaluation metric.

Compared with the F1-score, *PR-AUC is more appropriate for imbalanced datasets because it evaluates model performance across all possible classification thresholds. In contrast, the F1-score depends on a single threshold* and was therefore used later during threshold optimization rather than during model selection.

From the baseline analysis, four models were selected for further analysis:

- Random Forest
- XGBoost
- LightGBM
- CatBoost

5.11 Hyperparameter Optimization

To improve predictive performance, model hyperparameters were optimized using **Optuna**.

The optimization process involved:

- Bayesian hyperparameter optimization.
- Five-fold Stratified Cross-Validation.
- Precision-Recall Area Under the Curve (PR-AUC) as the optimization objective.

The optimized hyperparameters were subsequently used to train the final models.

To save computational power, only **10 trials** were conducted for each model.

5.12 Model Interpretation

To improve model transparency and interpretability, **Explainable Artificial Intelligence (XAI)** techniques were employed.

The interpretation process included:

- Feature Importance.
- SHAP (SHapley Additive exPlanations) Summary Plot.
- SHAP Feature Importance Plot.

These analyses identified the most influential financial indicators contributing to financial distress prediction.

5.13 Threshold Optimization

The default classification threshold of **0.50** was further optimized.

The optimal threshold was selected by:

- Evaluating Precision and Recall across different thresholds.
- Selecting the threshold that maximized the F1-score.

This step improved the practical usability of the prediction model.

5.14 Deployment Demonstration

To demonstrate the practical applicability of the developed framework:

- The final optimized model was applied to the latest available financial statements.
- Financial distress probabilities were estimated for the subsequent financial year.
- An **Early Warning Dashboard** was developed to identify companies with elevated financial distress risk.

Chapter 6

Data Analysis and Results

6.1 Dataset Overview

The dataset used in this study was constructed from publicly available financial statements of Indian publicly traded companies. Historical financial information was collected for multiple financial years, resulting in a panel dataset containing company-year observations. The dataset includes financial variables obtained from the balance sheet, profit and loss statement, and cash flow statement, along with engineered financial ratios used for predictive modelling.

After data collection and preprocessing, the dataset contained **4,373 company-year observations** representing **416 unique companies**. At this stage, the dataset had undergone duplicate removal, data cleaning, feature engineering, and missing value treatment, making it suitable for exploratory analysis and target variable construction.

Since the objective of this study was to develop an **Early Warning System**, the target variable (*FutureDistress*) was created by shifting the financial distress label one year ahead. As a result, the last available observation for each company could not be assigned a future distress status and was excluded from the modelling dataset. Consequently, the final dataset used for machine learning consisted of **3,957 company-year observations** representing **415 companies**.

Table 6.1 summarizes the characteristics of the dataset before and after the construction of the target variable.

Table 6.1: Dataset Summary

Characteristic	Before Target Creation	Final Dataset
Company-Year Observations	4,373	3,957
Unique Companies	416	415
Target Variable	Not Available	FutureDistress
Dataset Type	Panel Data	Panel Data

6.2 Data Quality Assessment

Before conducting the predictive modelling, the quality of the dataset was assessed to identify potential issues such as duplicate observations and missing values. Ensuring high data quality is essential for developing reliable machine learning models and improving prediction accuracy.

6.2.1 Duplicate Analysis

The collected dataset was first examined for duplicate observations. Duplicate records mainly arose during the web scraping process because financial information for some companies was retrieved multiple times.

After identifying duplicate entries, the dataset was carefully reviewed and duplicate observations were removed while retaining the correct financial records. This process reduced redundancy and ensured that each company-year observation was represented only once in the final dataset.

6.2.2 Missing Value Analysis

The distribution of missing values across the financial variables was examined after duplicate removal and preprocessing. Figure 6.1 presents the percentage of missing values for each variable.

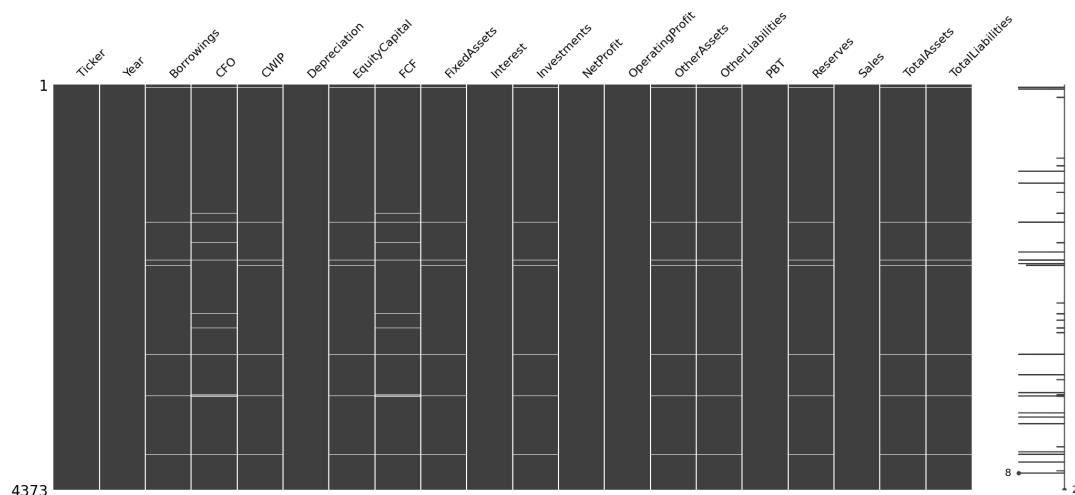


Figure 6.1: Missing data pattern across financial variables in the preprocessed dataset (4,373 company-year observations).

The distribution of missing values for each variable in the processed dataset is shown in Figure 6.1. Most financial variables contained very few missing observations, while several variables were completely available.

Variables such as **CFO**, **FCF**, **Interest**, **Operating Profit**, **Other Assets**, **Reserves**, **Sales**, **Total Assets**, and **Total Liabilities** contained only a small number of missing values. The variables **Ticker** and **Year** contained no missing observations.

6.3 Target Variable Analysis

Since the objective of this study is to develop an Early Warning System, a future financial distress label was constructed using accounting-based distress criteria.

Figure 6.2 illustrates the distribution of healthy and distressed observations in the final modelling dataset.

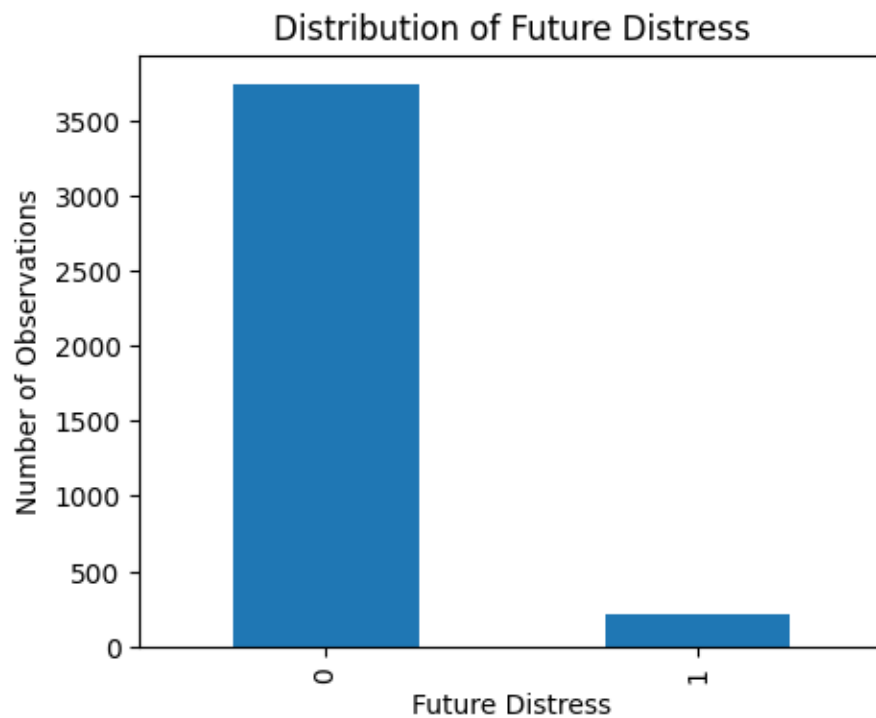


Figure 6.2: Distribution of the target variable (*FutureDistress*).

The dataset is **highly imbalanced**, with healthy observations substantially outnumbering financially distressed observations. This imbalance justifies the use of **SMO-TENC** during model training to improve the model's ability to identify financially distressed firms.

The sector-wise distribution of financially distressed companies is presented in Figure 6.3.

...	Companies	Observations	Distressed	DistressRate
Sector				
Communication Services	13	130	17	0.130769
Real Estate	11	114	12	0.105263
Financial Services	30	275	27	0.098182
Technology	39	367	27	0.073569
Industrials	83	839	52	0.061979
Consumer Cyclical	66	575	32	0.055652
Consumer Defensive	27	257	9	0.035019
Healthcare	50	455	14	0.030769
Basic Materials	62	603	18	0.029851
Utilities	20	195	4	0.020513
Energy	14	147	1	0.006803

Figure 6.3: Sector-wise distribution of financially distressed companies.

The distribution indicates that financial distress is present across multiple sectors, although its frequency varies considerably. This observation highlights the importance of developing a generalized prediction model capable of identifying financial distress irrespective of industry classification.

6.4 Exploratory Data Analysis

Exploratory data analysis was conducted to investigate the characteristics of the financial variables and identify underlying patterns before feature selection and model development.

6.4.1 Exploratory Analysis of Raw Financial Variables

In the first stage of the exploratory analysis, the raw financial statement variables were examined. These included financial measures such as Sales, Total Assets, Borrowings, Net Profit, Operating Profit, Cash Flow from Operations (CFO), Free Cash Flow (FCF), Interest Expense, and Equity Capital.

Descriptive analysis was performed to understand the variation in these financial variables across different company-year observations.

Figure 6.4 illustrates the Spearman correlation matrix of the raw financial statement variables.

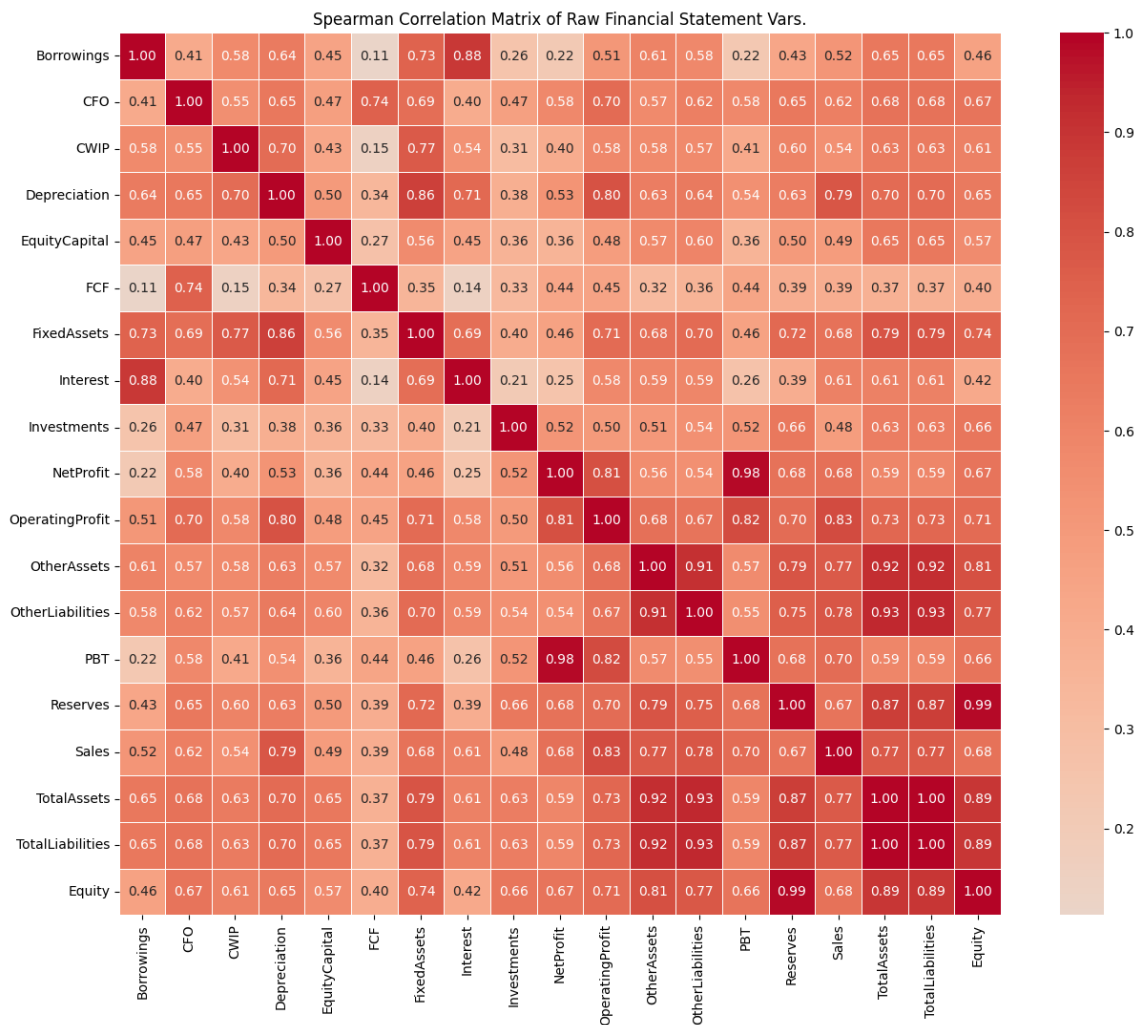


Figure 6.4: Spearman correlation matrix of the raw financial statement variables.

The correlation matrix indicates that several financial variables exhibit strong positive correlations. Larger firms generally report higher values across multiple financial measures, resulting in substantial multicollinearity among the raw variables.

Figure 6.5 presents the descriptive statistics of the raw financial variables together with the engineered variables.

***	count	mean	std	min	25%	50%	75%	max
Year	3957.0	2020.259287	3.294677	2005.000000	2018.000000	2021.000000	2023.000000	2.025000e+03
Borrowings	3957.0	9143.757139	30537.363362	0.000000	142.000000	821.000000	3679.000000	4.516640e+05
CFO	3957.0	2606.792014	9051.487684	-65502.000000	166.000000	576.000000	1625.000000	1.787030e+05
CWIP	3957.0	2578.561284	14656.659279	0.000000	19.000000	111.000000	594.000000	3.388550e+05
Depreciation	3957.0	667.968231	2393.663973	0.000000	38.000000	105.000000	328.000000	3.309100e+04
EquityCapital	3957.0	547.011635	2286.192832	0.000000	29.000000	78.000000	280.000000	7.139300e+04
FCF	3957.0	1023.968410	5935.782344	-112909.000000	-13.000000	233.000000	834.000000	8.676500e+04
FixedAssets	3957.0	11580.790751	41164.577660	0.000000	595.000000	1843.000000	6221.000000	9.993930e+05
Interest	3957.0	449.486447	1474.219066	-14.000000	10.000000	46.000000	197.000000	2.202700e+04
Investments	3957.0	12399.247157	172335.566767	0.000000	37.000000	315.000000	1491.000000	5.298429e+06
NetProfit	3957.0	1060.911946	3997.615324	-73878.000000	81.000000	264.000000	769.000000	6.784500e+04
OperatingProfit	3957.0	2284.053715	6585.413654	-18479.000000	188.000000	506.000000	1469.000000	1.085810e+05
OtherAssets	3957.0	12316.484963	33038.554772	0.000000	1254.000000	3235.000000	8683.000000	5.010590e+05
OtherLiabilities	3957.0	18112.333586	184912.269638	0.000000	622.000000	1799.000000	5524.000000	5.533342e+06
PBT	3957.0	1455.103232	5047.762077	-61797.000000	110.000000	355.000000	1018.000000	8.381500e+04
Reserves	3957.0	11072.697245	36519.786074	-154287.000000	1265.000000	3025.000000	7968.000000	8.296680e+05
Sales	3957.0	15061.300933	48333.912282	0.000000	1241.000000	3280.000000	9074.000000	7.236060e+05
TotalAssets	3957.0	38879.592621	206684.062004	0.000000	2819.000000	7360.000000	19374.000000	5.660568e+06
TotalLiabilities	3957.0	38879.592621	206684.062004	0.000000	2819.000000	7360.000000	19374.000000	5.660568e+06
Equity	3957.0	11619.708880	37006.323991	-104167.000000	1378.000000	3247.000000	8498.000000	8.432000e+05
DebtAssets	3954.0	0.217770	0.232132	0.000000	0.038884	0.171377	0.343821	6.448845e+00
SectorThreshold	3957.0	0.437798	0.109866	0.000000	0.357221	0.458191	0.496377	8.306011e-01
InterestCoverage	3767.0	42.277412	190.388549	-1799.000000	3.008263	8.520548	29.214777	4.846000e+03
ROA	3957.0	0.053887	0.108908	-4.396040	0.015437	0.038972	0.087455	9.818623e-01
ROE	3957.0	0.092208	0.949196	-52.306306	0.037882	0.096302	0.176944	1.088696e+01
OperatingMargin	3957.0	0.109558	4.449741	-277.000000	0.105386	0.169339	0.253550	1.040000e+00
NetMargin	3957.0	0.082446	3.329319	-199.000000	0.039940	0.086335	0.154365	5.600000e+01
DebtEquity	3957.0	0.860577	8.615598	-66.798817	0.051302	0.307866	0.890041	3.325938e+02
EquityRatio	3957.0	0.488187	0.268374	-5.825050	0.323053	0.500133	0.674222	9.979265e-01
CFOtoAssets	3957.0	0.103984	0.428402	-0.872558	0.047998	0.096304	0.143906	2.384000e+01
FCFtoAssets	3957.0	0.046762	0.192985	-0.960317	-0.003176	0.045167	0.095182	9.360000e+00
AssetTurnover	3957.0	0.654158	0.564582	0.000000	0.236289	0.521404	0.920046	6.721233e+00
FixedAssetRatio	3957.0	0.342460	0.208340	0.000000	0.177918	0.326363	0.493471	9.200313e-01
CWIPRatio	3957.0	0.043932	0.066943	0.000000	0.004173	0.018379	0.053751	5.734320e-01
InvestmentRatio	3957.0	0.125480	0.194411	0.000000	0.007785	0.044154	0.151443	9.966411e-01

Figure 6.5: Descriptive statistics of the raw financial statement and engineered variables.

Figure 6.5 shows considerable variability among the raw financial variables due to differences in firm size. Variables such as Cash Flow from Operations (CFO), Free Cash Flow (FCF), Net Profit, and Operating Profit exhibit substantial dispersion, highlighting the need for financial ratio construction before model development.

6.4.2 Exploratory Analysis of Engineered Financial Ratios

To overcome the limitations associated with absolute financial measures, a set of engineered financial ratios was constructed. These ratios normalize financial information relative to assets, equity, sales, or borrowings, making companies of different sizes more comparable.

Figure 6.6 presents the Spearman correlation matrix of the engineered financial ratios.

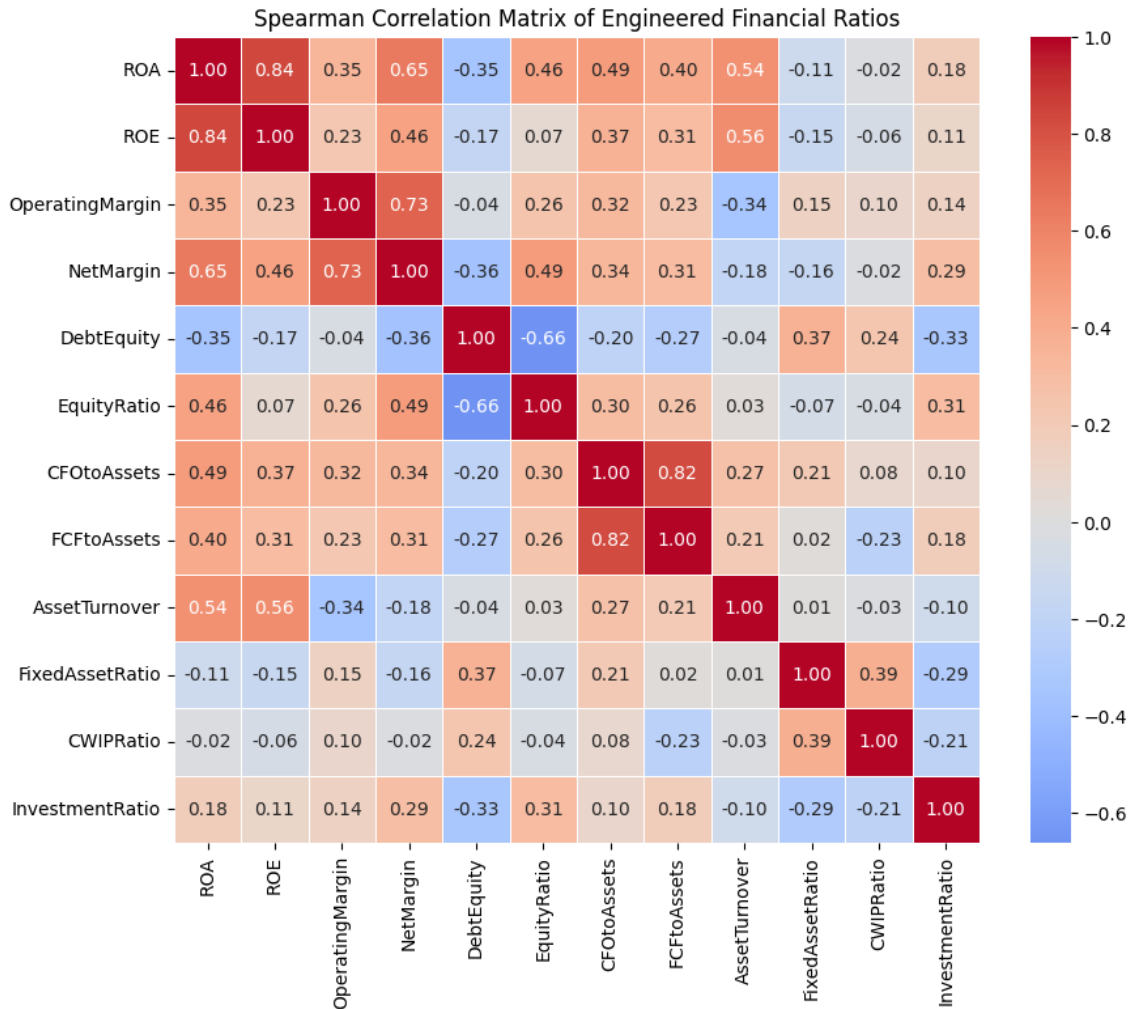


Figure 6.6: Spearman correlation matrix of the engineered financial ratios.

Compared with the raw financial variables, the engineered ratios exhibit substantially lower correlations, indicating that feature engineering successfully reduced the influence of firm size while preserving economically meaningful financial relationships. These engineered variables were subsequently used for statistical feature selection and machine learning model development.

Figure 6.5 gives an overview of the descriptive statistics for the constructed financial ratios in the current study. The profitability ratio statistics, such as ROA and ROE, have positive medians; this means that the vast majority of the firms were able to earn

profits during the time span covered by the study. Nonetheless, the presence of very low minima is indicative of the presence of financially troubled firms earning substantial losses. The leverage ratio statistics, like Debt/Equity and Equity Ratio, indicate large dispersions, implying that there were significant differences in terms of capital structure between firms.

6.5 Feature Selection

Selecting an appropriate set of predictor variables is a critical step in developing a reliable financial distress prediction model. Including irrelevant or highly correlated features may reduce model interpretability, introduce multicollinearity, and adversely affect predictive performance. Therefore, a multi-stage feature selection framework was adopted to identify the most informative financial indicators while maintaining statistical robustness and financial interpretability.

The feature selection process combined statistical hypothesis testing, correlation analysis, **Variance Inflation Factor (VIF)** analysis, and validation using **LASSO Logistic Regression**. Rather than relying on a single technique, these complementary approaches were applied sequentially to obtain a stable and interpretable set of predictors for subsequent machine learning model development.

Mann–Whitney U Test

Variables with p -values greater than 0.05 were excluded, while statistically significant variables were retained for further correlation and multicollinearity analysis.

Correlation Filtering and VIF Analysis

Highly correlated and multicollinear variables were removed based on Spearman correlation and VIF analysis (removing variables with high VIF values), resulting in a stable and non-redundant feature set for model development.

Final Selected Features

Table 6.2: Final Selected Features Used for Model Development

Feature	Feature	Feature
NetProfit	OperatingProfit	ROA
CFO	FCF	CFOtoAssets
Interest	DebtAssets	EquityRatio
InterestCoverage	FixedAssetRatio	CWIPRatio
Depreciation	OtherAssets	Sales
AssetTurnover	OtherLiabilities	Sector
Year		

A total of **19 features** were retained for subsequent model development.

Validation Using LASSO Logistic Regression

To verify the reliability of the selected features, **LASSO Logistic Regression** was used as an embedded feature selection method. Unlike the statistical feature selection framework, which combines significance testing and multicollinearity analysis, LASSO selects features by shrinking the coefficients of less important variables toward zero.

The features selected using LASSO were compared with those obtained through the statistical feature selection framework comprising the Mann–Whitney U test, correlation filtering, and VIF analysis.

- Result

Most of the predictor variables identified using the statistical framework were also selected by LASSO, except **Debt-to-Assets Ratio** and **CWIP Ratio**.

Since the statistically selected feature set was financially justifiable and largely consistent with the LASSO output, it (ones obtained from statistical framework) was retained for subsequent model development.

lasso_results				
...	Feature	Coefficient	Absolute	Selected
2	ROA	-4.323166	4.323166	True
8	EquityRatio	-2.687919	2.687919	True
5	CFOtoAssets	-0.391003	0.391003	True
10	FixedAssetRatio	-0.366791	0.366791	True
15	AssetTurnover	-0.282336	0.282336	True
9	InterestCoverage	-0.049486	0.049486	True
6	Interest	0.000473	0.000473	True
17	Year	-0.000425	0.000425	True
1	OperatingProfit	-0.000306	0.000306	True
3	CFO	-0.000253	0.000253	True
0	NetProfit	-0.000247	0.000247	True
16	OtherLiabilities	-0.000083	0.000083	True
12	Depreciation	0.000067	0.000067	True
13	OtherAssets	0.000028	0.000028	True
14	Sales	0.000018	0.000018	True
4	FCF	-0.000002	0.000002	True
7	DebtAssets	0.000000	0.000000	False
11	CWIPRatio	0.000000	0.000000	False

Figure 6.7: LASSO regression results compared with the statistically selected features.

6.6 Performance of Machine Learning Models

Introduction

The models considered for evaluation included Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, and CatBoost. Model performance was evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, and PR-AUC.

Due to the severe class imbalance in the dataset (5.38% distressed observations), **PR-AUC** was chosen as one of the primary evaluation metrics. **Unlike ROC-AUC, which incorporates the true negative rate, PR-AUC focuses only on the precision-recall relationship.** Consequently, ROC-AUC may remain high even when the model performs poorly on the minority class.

Similarly, **unlike the F1-score, which summarizes model performance at a single classification threshold, PR-AUC evaluates the precision-recall trade-off across all possible thresholds.** Therefore, since the primary objective of this study is to identify financially distressed firms, **PR-AUC** was considered the most appropriate evaluation metric.

6.6.1 Baseline Model Performance

comparison							
...	Model	Accuracy	Precision	Recall	F1	ROC_AUC	PR_AUC
0	CatBoost	0.963666	0.563218	0.890909	0.690141	0.986757	0.734025
1	LightGBM	0.966969	0.594937	0.854545	0.701493	0.986993	0.733405
2	XGBoost	0.968621	0.610390	0.854545	0.712121	0.986482	0.710622
3	Random Forest	0.970273	0.630137	0.836364	0.718750	0.985624	0.710158
4	Logistic Regression	0.931462	0.375000	0.763636	0.502994	0.938157	0.627659
5	Decision Tree	0.943848	0.405797	0.509091	0.451613	0.736812	0.228883

Figure 6.8: Performance Comparison of Baseline Models

The ensemble learning models consistently outperformed the individual classifiers across all evaluation metrics using the default model settings. CatBoost achieved the highest PR-AUC (0.734), while Random Forest recorded the highest Accuracy (97.03%) and F1-score (0.719). Overall, the ensemble models demonstrated superior capability in identifying financially distressed firms.

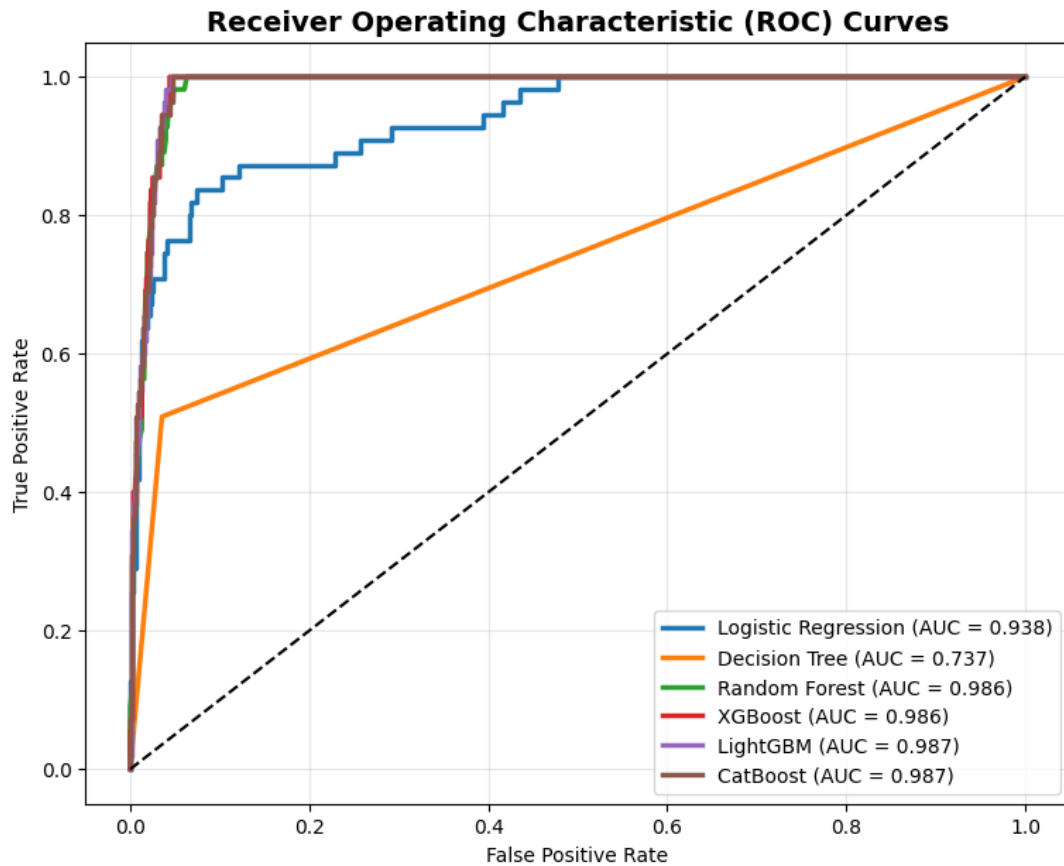


Figure 6.9: ROC Curves of Baseline Models

The ROC curves show that the ensemble models achieved excellent class discrimination, with ROC-AUC values close to 0.99. CatBoost and LightGBM exhibited the best overall discrimination, whereas Logistic Regression and Decision Tree showed comparatively weaker performance.

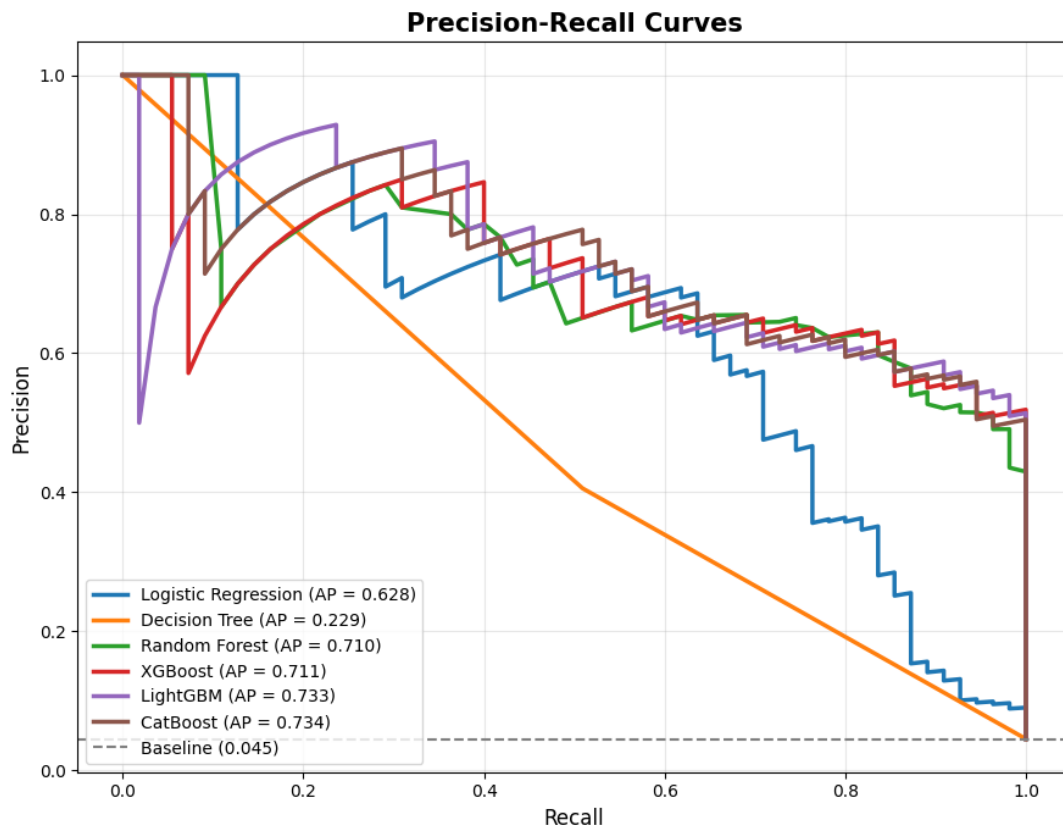


Figure 6.10: Precision-Recall Curves of Baseline Models

The Precision-Recall curves provide a more informative comparison under class imbalance. CatBoost and LightGBM achieved the highest PR-AUC values, indicating a better balance between Precision and Recall when identifying financially distressed firms, while the Decision Tree model performed the poorest.

6.6.2 Hyperparameter Tuning

Hyperparameter tuning of the top-performing ensemble learning algorithms, namely **Random Forest**, **XGBoost**, **LightGBM**, and **CatBoost**, was performed using **Optuna** to obtain the optimal set of hyperparameters for each algorithm.

The optimized hyperparameters improved the learning configuration of each model while maintaining computational efficiency.

Table 6.3: Optimized Hyperparameters of the Ensemble Models

Model	Optimized Hyperparameters
Random Forest	n_estimators = 191, max_depth = 13, min_samples_split = 3
XGBoost	n_estimators = 111, max_depth = 8, learning_rate = 0.1242
LightGBM	n_estimators = 242, num_leaves = 21, learning_rate = 0.1943
CatBoost	iterations = 175, depth = 8, learning_rate = 0.1491

The optimized hyperparameters were subsequently used to train the final models.

6.6.3 Performance of Optimized Models

The performance of the optimized ensemble learning models was compared with their corresponding baseline models to evaluate the effectiveness of hyperparameter tuning.

Model_Optimized	Accuracy_Optimized	...	Recall_Optimized	F1_Optimized	ROC_AUC_Optimized	PR_AUC_Optimized	Δ Accuracy	Δ Precision	Δ Recall	Δ F1	Δ ROC_AUC	Δ PR_AUC
Optimized Random Forest	0.9703	...	0.8545	0.7231	0.9875	0.7522	0.0000	-0.0035	0.0182	0.0043	0.0019	0.0421
Optimized XGBoost	0.9653	...	0.8000	0.6769	0.9866	0.7267	-0.0033	-0.0237	-0.0545	-0.0352	0.0001	0.0161
Optimized LightGBM	0.9727	...	0.8909	0.7481	0.9864	0.7052	0.0058	0.0498	0.0364	0.0466	-0.0006	-0.0282
Optimized CatBoost	0.9604	...	0.8909	0.6712	0.9857	0.7054	-0.0033	-0.0248	0.0000	-0.0189	-0.0010	-0.0286

Figure 6.11: Performance Comparison of Baseline and Optimized Ensemble Models

- Hyperparameter optimization improved the performance of the Random Forest model, resulting in the highest PR-AUC (0.7522) among all optimized models.
- The XGBoost model showed a moderate improvement in PR-AUC, although its Recall and F1-score decreased slightly after optimization.
- LightGBM achieved improvements in Accuracy, Precision, Recall, and F1-score; however, its PR-AUC decreased marginally compared with the baseline model.
- The optimized CatBoost model maintained a high Recall but experienced a slight decline in Precision, F1-score, and PR-AUC after tuning.

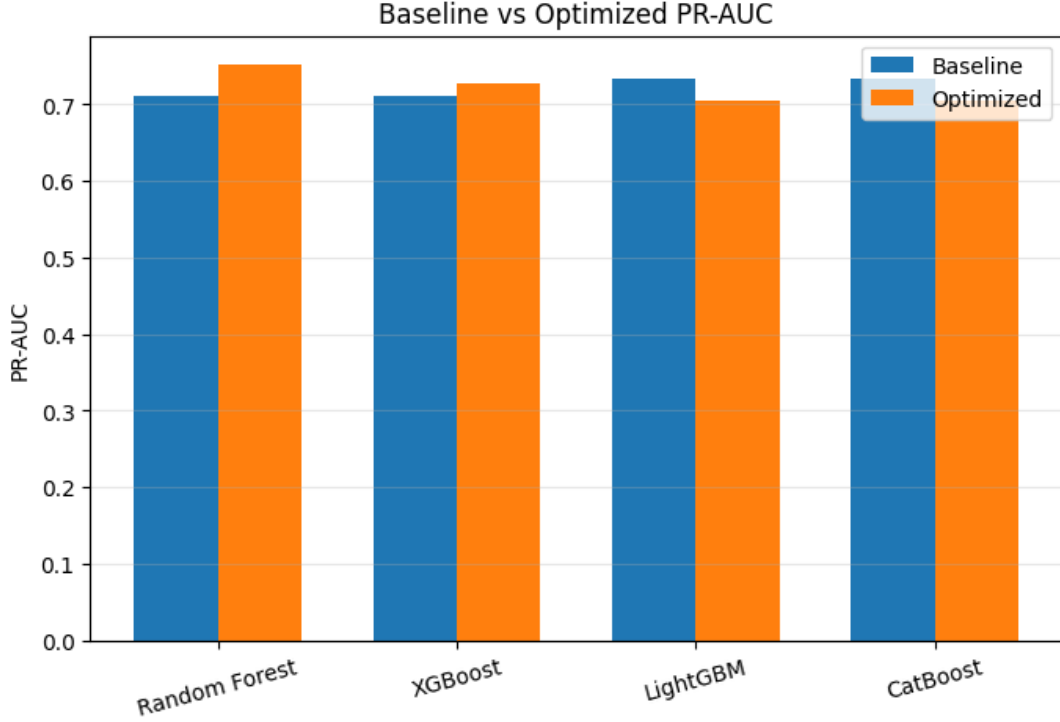


Figure 6.12: PR-AUC Comparison of Baseline and Optimized Ensemble Models

The **Random Forest** classifier showed the most significant improvement, achieving the highest **PR-AUC (0.7522)**. It also provided a comparatively better balance between Recall and F1-score among all the optimized models, making it the best-performing algorithm in this study.

6.6.4 Model Selection

According to the evaluation results of the optimized models, the **Random Forest** classifier was selected as the final model for the proposed Early Warning System.

Although all ensemble models achieved comparable ROC-AUC values, the Random Forest model recorded the highest **PR-AUC (0.7522)** and demonstrated superior performance in identifying financially distressed companies under severe class imbalance.

Furthermore, the model achieved an effective balance between Precision, Recall, and F1-score while maintaining high overall classification performance.

Therefore, the **Optimized Random Forest** model was selected for further analysis.

6.6.5 Interpretation of the Model via SHAP

To enhance the interpretability of the optimized Random Forest model, **SHapley Ad-ditive exPlanations (SHAP)** were applied to evaluate the impact of each financial variable on the model's prediction results.

Feature Importance

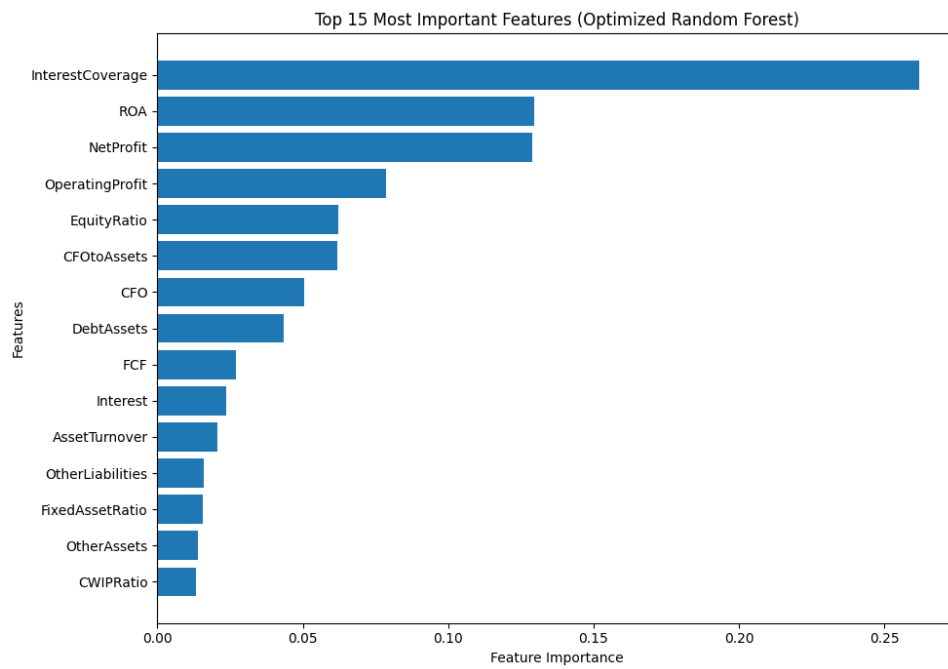


Figure 6.13: Feature Importance

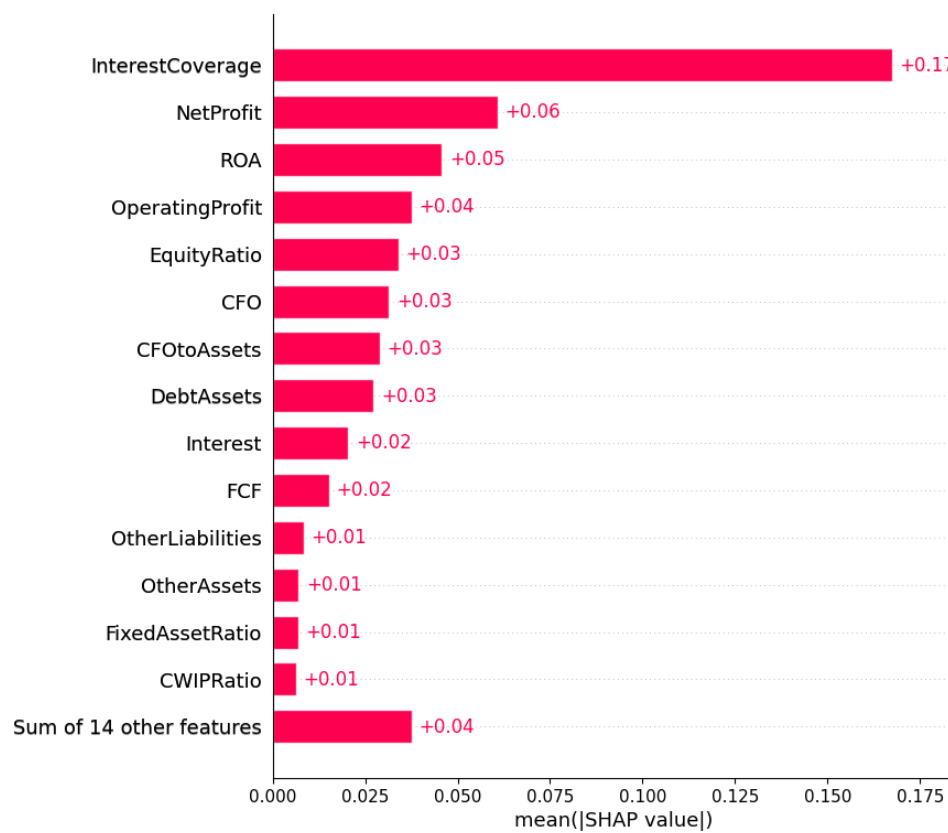


Figure 6.14: SHAP Feature Importance

Both analyses represent the feature importance obtained from the optimized Random Forest model. **Interest Coverage** emerged as the most influential predictor, followed by **Return on Assets (ROA)**, **Net Profit**, **Operating Profit**, **Equity Ratio**, and **Cash Flow from Operations to Total Assets (CFO-to-Assets)**. These results indicate that profitability, debt-servicing capacity, and cash flow measures are the primary determinants of corporate financial distress.

SHAP Beeswarm Plot

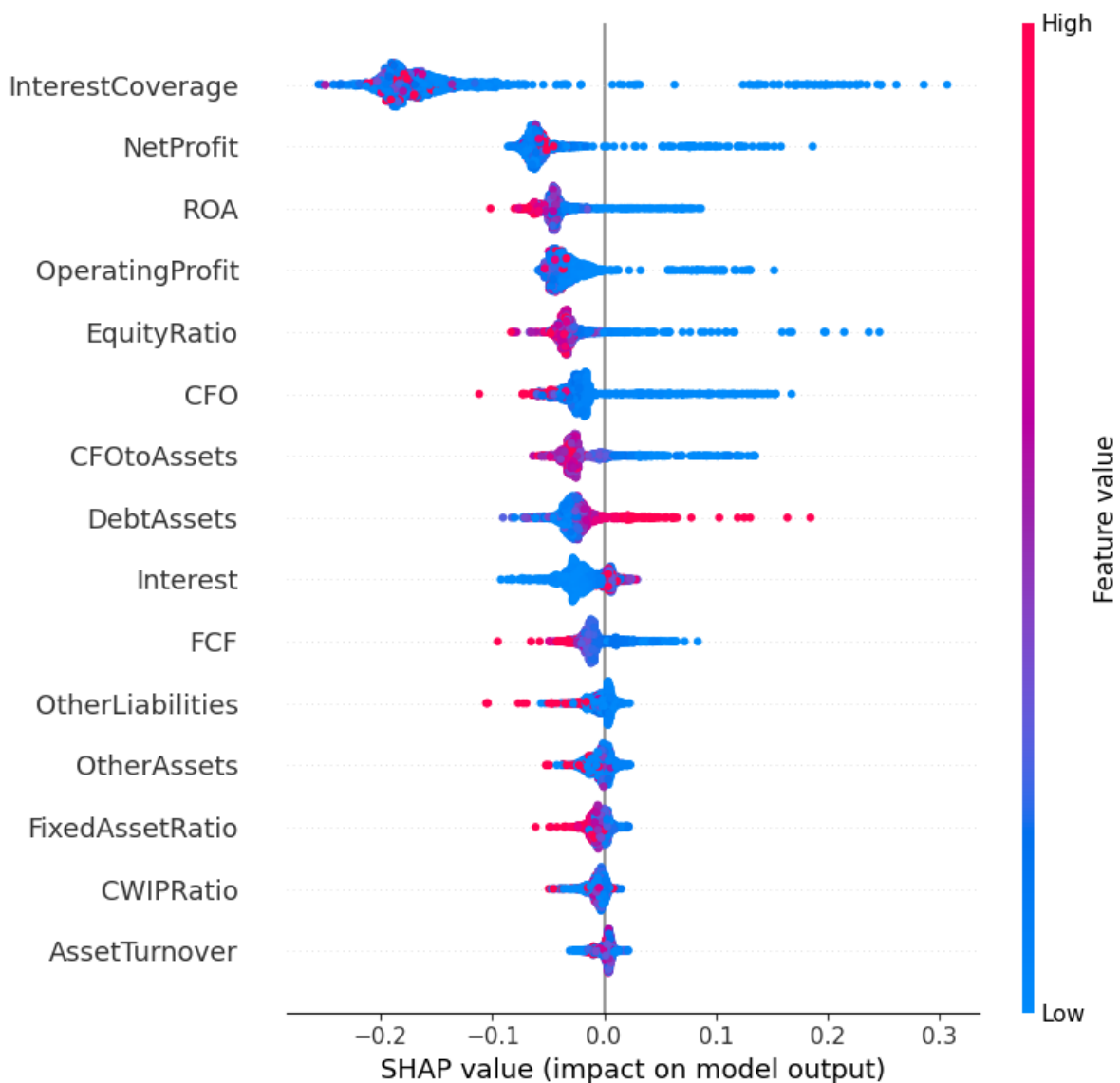


Figure 6.15: SHAP Summary (Beeswarm) Plot

Figure 6.15 presents the SHAP summary (beeswarm) plot, which illustrates both the importance and direction of each feature's contribution to the model's predictions.

- **Red** = High feature value
- **Blue** = Low feature value
- **Right (+SHAP)** = Increases the predicted probability of financial distress
- **Left (−SHAP)** = Decreases the predicted probability of financial distress

Taking **Interest Coverage** as an example:

- Low Interest Coverage (blue) appears on the right, indicating a higher predicted probability of financial distress.
- High Interest Coverage (red) appears on the left, indicating a lower predicted probability of financial distress.

The distribution of SHAP values indicates that low Interest Coverage, Net Profit, ROA, Operating Profit, CFO, CFO-to-Assets, and Equity Ratio generally contribute positively to the prediction of financial distress, whereas higher values of these indicators reduce the predicted risk.

Conversely, higher Debt-to-Assets ratios increase the predicted probability of financial distress, reflecting the adverse effect of excessive leverage on corporate financial stability.

In summary, the SHAP analysis demonstrates that the optimized Random Forest model relies on economically meaningful financial variables when making predictions. Profitability, debt-servicing ability, leverage, and cash flow indicators are the most influential determinants of financial distress.

6.6.6 Threshold Optimization

The optimized Random Forest model predicts the probability of future financial distress rather than directly assigning class labels. Instead of using the default classification threshold of **0.50**, threshold optimization was performed to identify the probability cutoff that maximizes the **F1-score**.

Since the dataset is imbalanced, the F1-score provides an appropriate balance between Precision and Recall, resulting in a more effective decision threshold for the proposed Early Warning System. It should also be noted that **PR–AUC is independent of the classification threshold**.

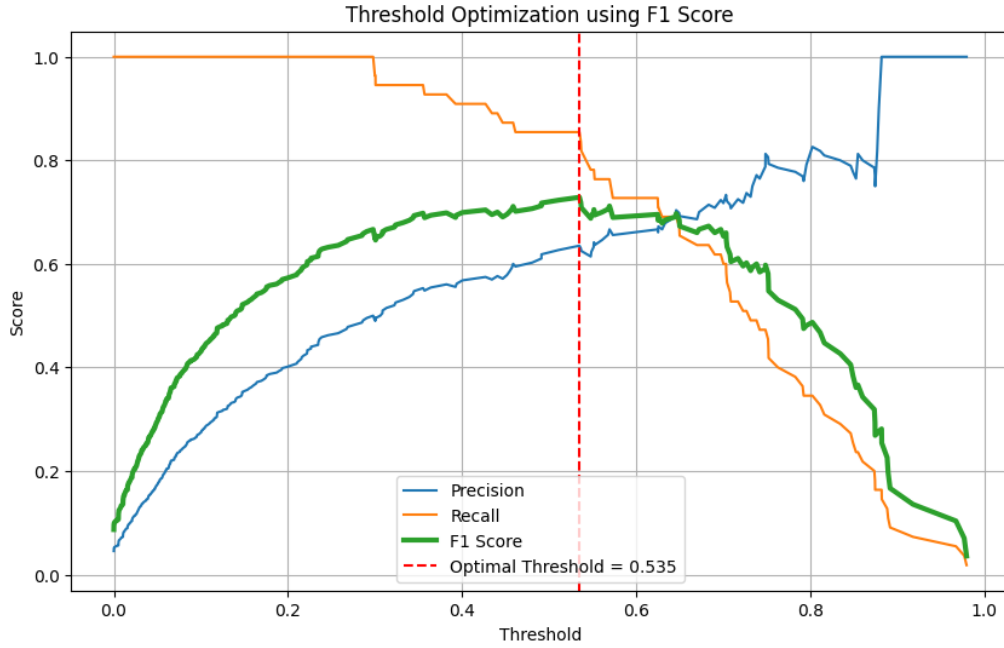


Figure 6.16: Threshold Optimization

The threshold optimization plot illustrates how Precision, Recall, and F1-score vary across different probability thresholds for the optimized Random Forest model.

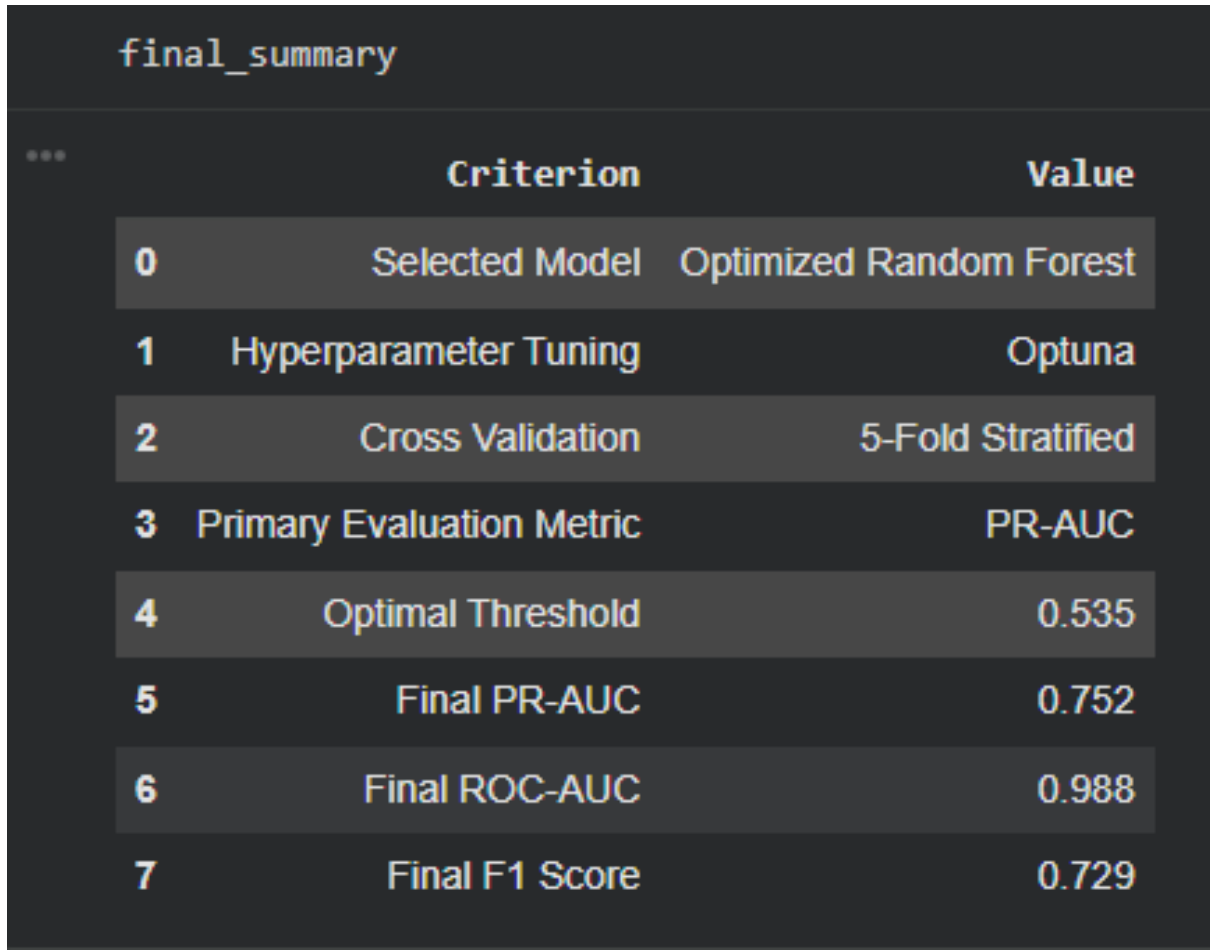
The maximum F1-score was achieved at a threshold of **0.535**, which was selected as the optimal decision boundary for classifying firms as financially distressed or healthy.

A threshold of **0.535** implies that a company is classified as financially distressed only if its predicted probability of future distress is at least **53.5%**. Otherwise, it is classified as financially healthy.

This result suggests that a slightly stricter cutoff of **53.5%** provides a better balance between correctly identifying distressed firms (Recall) and avoiding false alarms (Precision).

6.6.7 Final Summary

Figure 6.17 presents a snapshot of the final performance of the optimized Random Forest model.



```
final_summary
```

	Criterion	Value
0	Selected Model	Optimized Random Forest
1	Hyperparameter Tuning	Optuna
2	Cross Validation	5-Fold Stratified
3	Primary Evaluation Metric	PR-AUC
4	Optimal Threshold	0.535
5	Final PR-AUC	0.752
6	Final ROC-AUC	0.988
7	Final F1 Score	0.729

Figure 6.17: Final Summary of the Optimized Random Forest Model

The confusion matrix shown in Figure 6.18 summarizes the classification performance of the optimized Random Forest model using the selected threshold of **0.535**.

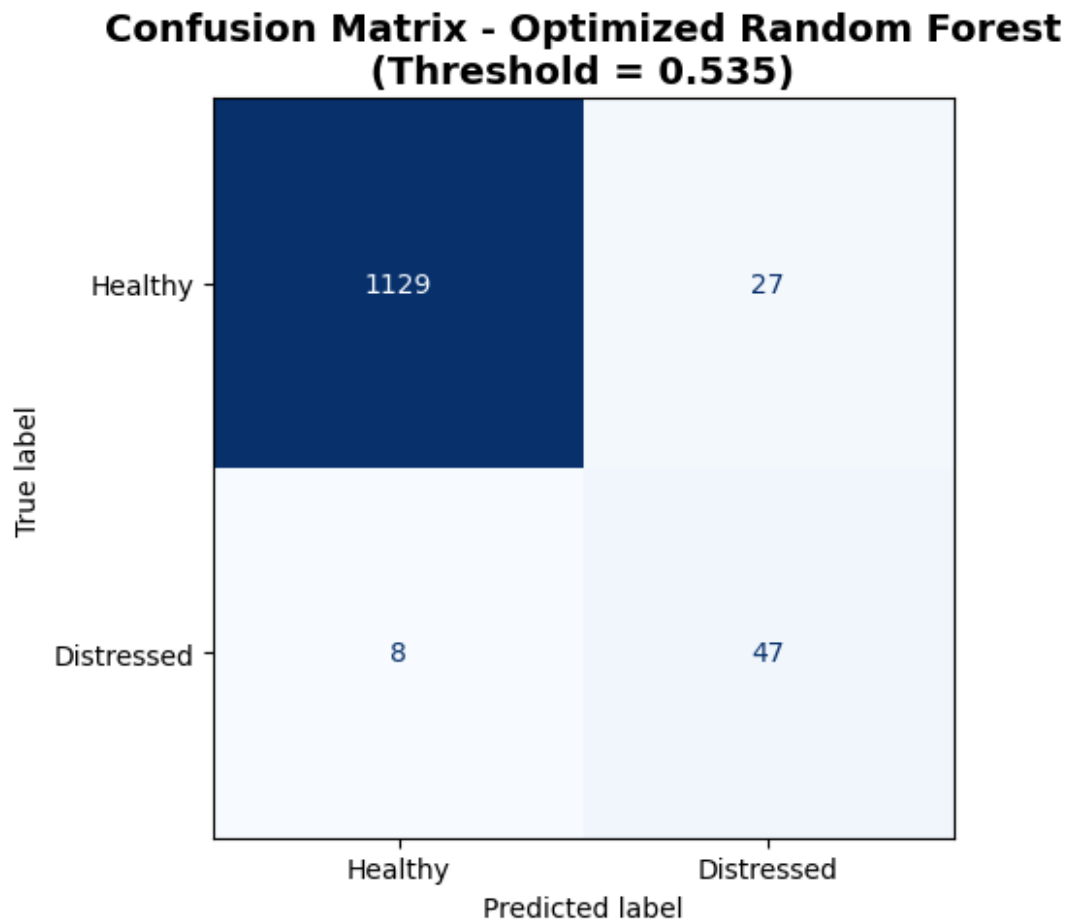


Figure 6.18: Confusion Matrix of the Optimized Random Forest Model

- **True Negatives (1,129):** Healthy companies correctly identified as financially healthy.
- **True Positives (47):** Financially distressed companies correctly identified as distressed, representing successful early warning predictions.
- **False Positives (27):** Healthy companies incorrectly flagged as distressed, resulting in unnecessary caution or further investigation (false alarms).
- **False Negatives (8):** Financially distressed companies incorrectly classified as healthy. These represent missed distress cases and are the most critical classification errors, as they may prevent timely intervention.

6.7 Early Warning Dashboard

To demonstrate the practical application of the proposed **Early Warning System**, two visual dashboards were developed.

Early Warning Dashboard: Top 10 Companies by Predicted Financial Distress Risk

Ticker	Year	Actual Status	Predicted Status	Distress Probability (%)	Risk Level
SWIGGY	2024	Distressed	Distressed	97.99	Very High
URBANCO	2023	Distressed	Distressed	97.71	Very High
SWIGGY	2023	Distressed	Distressed	96.76	Very High
SWIGGY	2025	Distressed	Distressed	91.83	Very High
IDEA	2025	Distressed	Distressed	89.2	Very High
OLAELEC	2025	Distressed	Distressed	88.97	Very High
360ONE	2025	Distressed	Distressed	88.86	Very High
ONESOURCE	2023	Distressed	Distressed	88.24	Very High
ITI	2024	Distressed	Distressed	88.22	Very High
GLENMARK	2023	Healthy	Distressed	87.9	Very High

Figure 6.19: Early Warning Dashboard – Top Companies by Predicted Financial Distress Risk

The dashboard shown in Figure 6.19 ranks companies according to their predicted probability of financial distress. For each company, the dashboard reports the ticker, financial year, actual financial status, predicted status, estimated distress probability, and corresponding risk level. The results show that the majority of the highest-risk companies were correctly identified as financially distressed, demonstrating the model's effectiveness in prioritizing firms with elevated financial risk.

To demonstrate the predictive capability of the proposed Early Warning System, the optimized Random Forest model was applied to the 2025 financial statements to predict the probability of financial distress in 2026.

**Prediction of Financial Distress for 2026
(Using Financial Statements of 2025)**

Ticker	Financial Year	Predicted Distress (2026)	Distress Probability (%)
SWIGGY	2025	Distressed	91.83
IDEA	2025	Distressed	89.2
OLAELEC	2025	Distressed	88.97
360ONE	2025	Distressed	88.86
TEJASNET	2025	Distressed	83.47
ABREL	2025	Distressed	79.15
PIRAMALFIN	2025	Distressed	78.3
GODREJPROP	2025	Distressed	74.86
SIGNATURE	2025	Distressed	73.37
GODREJIND	2025	Distressed	72.69
BEML	2025	Distressed	72.31
ITI	2025	Distressed	70.38
URBANCO	2025	Distressed	70.35
MEESHO	2025	Distressed	68.35
KIRLOSENG	2025	Distressed	62.54
ONESOURCE	2025	Distressed	56.79
IRCON	2025	Distressed	54.87
ANGELONE	2025	Healthy	46.15
PPLPHARMA	2025	Healthy	44.68
GMRAIRPORT	2025	Healthy	42.71

Figure 6.20: Prediction of Financial Distress for 2026 Using 2025 Financial Statements

The dashboard identifies firms that are likely to experience financial distress and reports the corresponding predicted probabilities. These results demonstrate that the proposed system can identify potential financial problems in advance, highlighting its usefulness as an effective early warning tool.

Chapter 7

Conclusion

7.1 Conclusions

This study successfully developed an **Early Warning System (EWS)** for predicting the future financial distress of Indian publicly listed companies using accounting-based financial indicators and machine learning techniques. Based on the results obtained, the following conclusions can be drawn:

1. Machine learning techniques can effectively predict financial distress using accounting-based financial information. The developed framework successfully distinguished between healthy and financially distressed firms, demonstrating that financial statement variables provide sufficient predictive information for early distress identification.
2. The optimized Random Forest model achieved the best overall predictive performance. Among the evaluated machine learning algorithms, the optimized Random Forest classifier recorded the highest PR-AUC (0.752) while also achieving a ROC-AUC of 0.988 and an F1-score of 0.729. Given the severe class imbalance in the dataset, PR-AUC was considered the primary evaluation metric, supporting the selection of Random Forest as the final prediction model.
3. Feature selection improved model robustness and interpretability. The combination of the Mann-Whitney U test, Spearman correlation analysis, Variance Inflation Factor (VIF) analysis, and LASSO validation resulted in a stable and interpretable set of financial predictors. The strong agreement between the statistical feature selection framework and LASSO further validated the robustness of the selected variables.
4. Model interpretation confirmed the financial relevance of the selected predictors. SHapley Additive exPlanations (SHAP) analysis identified Interest Coverage, Return on Assets (ROA), Net Profit, Operating Profit, and Equity Ratio as the most

influential variables affecting financial distress prediction. These findings are consistent with established financial theory, demonstrating that the model relies on economically meaningful indicators rather than spurious patterns.

5. Threshold optimization enhanced the practical applicability of the model. Optimizing the classification threshold to 0.535 improved the balance between Precision and Recall, reducing the likelihood of missing financially distressed firms while maintaining satisfactory predictive performance.
6. The proposed framework can serve as a practical decision-support tool. The developed Early Warning Dashboard demonstrated how the optimized model can be used to rank companies according to financial distress risk and generate forward-looking predictions using the latest available financial statements. This enables investors, lenders, regulators, and corporate managers to identify financially vulnerable firms and undertake timely risk management actions.

Overall, the study demonstrates that integrating financial statement analysis with explainable machine learning provides an effective and interpretable framework for the early identification of corporate financial distress.

7.2 Future Scope

Although the proposed Early Warning System achieved strong predictive performance, several opportunities exist for further improvement:

- Incorporate market-based variables, such as stock returns, market capitalization, and price volatility, to complement accounting-based indicators.
- Include macroeconomic variables, including GDP growth, inflation, and interest rates, to capture the influence of broader economic conditions on corporate financial distress.
- Explore alternative machine learning and deep learning techniques, such as neural networks, transformer-based models, or graph neural networks, for potential improvements in predictive performance.
- Extend the study to include private companies or firms from other countries to evaluate the generalizability of the proposed framework.
- Develop a real-time Early Warning Dashboard integrated with automated financial data collection to support continuous monitoring and timely risk assessment.

Appendices

References

- Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). *Optuna: A Next-generation Hyperparameter Optimization Framework*. Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery & Data Mining.
- Altman, E. I. (1968). Financial Ratios, Discriminant Analysis and the Prediction of Corporate Bankruptcy. *The Journal of Finance*, 23(4), 589–609.
- Beaver, W. H. (1966). Financial Ratios as Predictors of Failure. *Journal of Accounting Research*, 4, 71–111.
- Dovilė Kuizinienė, Tomas Krilavičius, Robertas Damaševičius, & Rytis Maskeliūnas. *Systematic Review of Financial Distress Identification Using Artificial Intelligence Methods*.
- Farida Titik Kristanti, Dwi Fitrizal Salim, & Aditya Firman Ihsan. *Financial Distress Prediction in Emerging Markets by Using a Machine Learning Approach for Financial Sustainability*.
- Grewal, T. S. (n.d.). *Introduction to Accounting*. Sultan Chand & Sons.
- Klietlik, T., Valaskova, K., Nica, E., Durana, P., & Kovacova, M. (2024). Country-specific Financial Distress Prediction Using Decision Trees: Empirical Evidence from Slovakia and Croatia. *Journal of Risk and Financial Management*, 17(9), 432.
- Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems (NeurIPS)*.
- National Stock Exchange of India (NSE).
- Ohlson, J. A. (1980). Financial Ratios and the Probabilistic Prediction of Bankruptcy. *Journal of Accounting Research*, 18(1), 109–131.

- Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Screener.in.
- Towards Data Science. (2020). *Optimal Threshold for Imbalanced Classification*.
- Wall Street Prep. (n.d.). *Financial Restructuring: Common Causes of Financial Distress*.

GitHub Repository Link

<https://github.com/ashis-72/Early-Warning-System-Financial-Distress>